

Philosophical machines under incomplete knowledge: BOIS, SIMA, BORIS, substrate physiology, and accountable AI

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Author's Introduction

I was born in the Far East of Russia, in the Jewish Autonomous Region. I never thought that I would happen to make any kind of discovery. Especially since I never completed university: I faced a choice between earning money and studying, and I chose earning money.

However, my interest in philosophy led to the discovery described in the article below.

This discovery is formulated for firms because I was in a hurry to publish it. However, I sincerely believe that future researchers will find it very interesting for explaining the phenomena of life and consciousness.

This discovery is a closed cycle that feeds on attention and the unknown, and produces verified experience and increasing diversity under simultaneous pressure toward cooperation. A kind of philosophical machine. The exact domain of definition is not available to me; therefore, for publication I made a deliberate decision to restrict the domain narrowly and practically: the construction of business plans plus support for their execution. This is a theoretical work, and although tests are taking place right now, it has not yet gone beyond the scientific laboratory of a contemporary philosopher - his writing desk with a computer.

To avoid interfering with the creative work of ChatGPT, I provide here the sources I used. I read these books over the course of my life, and the ideas from these books form the basis of my discovery.

The translation from Russian was performed by ChatGPT using the protocol described here.

Sources indicated by the author and kept inside the author section:

The Bhagavad Gita.

M. K. Mamardashvili, Philosophical Readings. Moscow: Azbuka-Klassika, 2002. 832 pp. Includes: Introduction to Philosophy; Aesthetics of Thinking; Cartesian Meditations.

A. G. Tyslenko, Management. Organizational Structures of Management. Moscow: Alfa-Press, 2011. 320 pp.

F. A. Hayek, The Road to Serfdom.

Karl Marx, Capital, Volume 1.

Richard Dawkins, The Selfish Gene.

ISO 9001 [international quality management system standard] as the author's reading of John Boyd's ideas.

Note: this list is part of the author introduction and is not moved into the general reference list of the article.

Nonessential author notes

In subsequent work, my surname is given in a shortened form because, after full legal formalization in the United States, I intend to remove the local "ov" ending from the surname and keep only the indeclinable root, which is more convenient for the English-speaking world.

Disclosure of text generation and use of AI

This v4.6 preprint manuscript was prepared by updating the v4.5 core-aligned manuscript with the author-supplied preprint metadata: no ORCID, no institutional affiliation, self-funded work, no competing interests identified, and no public repository/materials link in this version. Generative AI was used as an assistant for synthesis, drafting, translation, formatting, and quality checks under the author's direction; AI is not an author. The human author remains responsible for claims, references, final edits, public posting decisions, and any later journal disclosures. The Author's Introduction above is preserved except for one explicit author-approved bibliographic correction: adding M. K. Mamardashvili as the author of Philosophical Readings.

Summary

AI governance often begins with outputs, risks, and values, but a deeper operational layer is decisive: the machine that defines objects, evidence, agency, risk, closure, and transition. This article updates the BOIS protocol under current BOIS/BORIS v2.6 by adding SIMA as the analytical layer and BORIS as the implementation layer. BOIS is treated as a transition-oriented philosophical machine for organized consequential activity under incomplete knowledge, not as a universal theory or field-validated product. SIMA reconstructs candidate machines from substrate physiology through operas, operome inventories, and operature correspondence. BORIS builds narrow domain implementations that must be tested by experience. The substrate rule M@S remains central: the same abstract machine, run on different relevant substrates, may preserve a high-level function while acquiring different implementation physiologies.

Keywords

philosophical machines; BOIS; SIMA; BORIS; incomplete knowledge; substrate physiology; operome; accountable AI; organized activity; protocols; agency; value interface

The missing operational layer

The practical question for AI-mediated work is no longer only whether a model can produce fluent or accurate sentences. The harder question is which architecture of reasoning is activated when a system answers, recommends, summarizes, escalates, or closes a task. A system may be locally accurate while importing a fragile ontology, an implicit theory of evidence, an unauthorized value judgment, or a premature sense of completion. These failures are not merely technical; they are philosophical failures made operational.

A philosophical machine is an inspectable architecture for inquiry and action. It specifies what kinds of entities are admitted, what counts as evidence, who owns agency, how purposes enter, what risks stop action, how a cycle is closed, and what is carried forward. This is not a claim that a machine is conscious or that software becomes a philosopher. It is a claim that systems can instantiate rule-governed conditions of reasoning, and that those conditions should be made visible before they act on the world.

The minimal design can be written as $M = \langle O, E, A, V, R, C, T \rangle$. O is ontology; E is epistemology; A is agency; V is the value interface; R is the risk model; C is the closure rule; and T is the transition rule. Different philosophical machines can use the same language model while producing different forms of inquiry. A medical triage assistant, a legal intake tool, a research assistant, a planning system, and a classroom tutor should not silently share one epistemology simply because they share a model. This frame draws on bounded rationality, theories of artificial systems, cybernetics, inquiry, and AI-governance literature (1-9, 23-31).

Table 1. Design axes for philosophical machines

Layer	Design question	Failure if hidden
Ontology	What entities, relations, and boundaries are admitted?	Ontology laundering: a convenient model is treated as reality.
Epistemology	What counts as evidence, hypothesis, interpretation, or verified result?	False certainty, endless skepticism, or evidence theater.
Agency	Who sets goals, authorizes action, revises, or stops?	Automation inherits authority invisibly.
Value interface	Where do purposes, constraints, and forbidden losses enter?	Values appear neutral because their source is hidden.
Risk model	Which errors are costly enough to stop or escalate?	Speed is optimized while harm is externalized.
Closure rule	When is a cycle complete?	A summary is mistaken for resolution.
Transition rule	What persists into the next cycle?	Context, uncertainty, rollback, and responsibility are lost.

BOIS as a protocol machine of incomplete knowledge

BOIS, in the author's working materials, is one philosophical machine within this broader class. Its canonical operational case remains the firm or economic organization, because that was the initial domain in which the cycle was articulated. In the current core, however, the word business in BOIS/BORIS is used in the broad project-internal sense of organized consequential activity: structured action with aims, roles, memory, costs, feedback, and repair. It should not be read only as commercial enterprise, and it should not be stretched into everything without protocol.

The current notation is BOIS(D,V,S). D is the domain of activity; V is the external value interface that supplies purposes, constraints, acceptable risk, time horizon, and forbidden losses; S is the substrate on which the machine is run. A transfer beyond firms is therefore not automatically valid. It becomes a marked research transfer or candidate implementation until D, V, S, evidence translation, risk controls, stop conditions, and experience tests are specified.

The minimal BOIS cycle is: released attention -> ignorance or domain of ignorance -> valuable unknown -> valuable question -> protocol -> next step -> execution -> experience -> feedback and selection -> verified result -> instruction -> physiology -> released attention. The cycle is not a metaphysics of everything. It is a reconstructive minimum for how organized activity can avoid collapsing unknowns, questions, protocols, steps, experience, verified results, and instructions into one another.

The key distinction is between protocol and instruction. A protocol is the form of work used when the situation is unknown, insufficiently tested, or expensive to get wrong. An instruction is the form of execution for what is already sufficiently verified in a defined domain. A verified result is therefore not eternal truth; it is a local and revisable outcome of protocol work. A next step is not knowledge; it is a hypothesis of state change. Automation is healthy only when it executes instructions inside their domain and returns out-of-domain cases to protocol rather than hiding them.

BOIS depends on an external value interface. It does not generate the subject's goals, legitimate those goals, or become the world it models. This dependency prevents a reasoning system from treating its own procedural fluency as moral authority.

Table 2. BOIS recovery minimum after v2.6 alignment

Element	Minimum formulation
Domain	The firm remains the canonical operational case; other domains require transfer protocol.
Business	In this project, business means organized consequential activity, not only commercial activity.
Incomplete knowledge	Complete knowledge of the activity, environment, future, and consequences is unattainable.
Value interface	Goals, constraints, acceptable risk, horizon, and forbidden losses enter from outside BOIS.
Next step	A next step is a hypothesis of state change, not knowledge or guarantee.
Protocol	A protocol handles what is unknown, unverified, or costly to get wrong.
Instruction	An instruction executes what is verified inside a defined domain and has stop signals.
Physiology	Physiology should release attention and return experience into improved routines.
SIMA	SIMA is the analytical layer that reconstructs candidate machines and risks from substrate physiology.
BORIS	BORIS is the implementation layer, not the BOIS core.
Transfer	Transfer beyond firms requires D, V, S, evidence translation, risk analysis, and experience tests.

SIMA as analytical layer: physiology, operates, and operates

Current BOIS/BORIS adds a second layer between abstract core and implementation: SIMA, the Substrate-Independent Machine Analyzer. SIMA is not another core and not a proof that BOIS is empirically valid. It is an analytical instrument. Its task is the inverse problem: given observed data about the physiology of a real object on a substrate, reconstruct candidate philosophical machines that could be operating there.

This inverse problem is underdetermined. A physiology can be compatible with more than one machine. SIMA therefore should not issue a single hidden diagnosis. It should produce candidate machines, maps of correspondence, confidence levels, rival explanations, and tests that could refute the reconstruction.

The intermediate unit is the oper: a minimal observable transition through which a machine distinguishes a situation, accepts evidence, distributes agency, applies a value or risk gate, changes state, writes memory, closes a micro-step, carries residue into the next transition, and pays a cost. The operome is the controlled term for the set of observed or candidate ops of an object X during a specified observation window W. The term is useful in extended reports but should not be forced into short analyses.

SIMA also replaces the earlier tempting metaphor of 'operature genetics' with operature correspondence. The point is not that organizations or AI workflows literally have genes. The point is that comparable machines can be described through partial correspondences among ops: evidence-channel to evidence-channel, memory-write to memory-write, closure-trigger to closure-trigger, repair to repair, and cost to cost. This makes cross-substrate comparison more formal without biologizing the analysis.

SIMA has two places in the cycle. Before a new BORIS is designed, SIMA analyzes the current object, its substrate, its operating machine, its operome, and its risks. After BORIS is implemented, SIMA audits whether the new physiology actually releases attention, preserves layer boundaries, reduces risk, and returns experience into improved instructions.

Table 3. BOIS, SIMA, and BORIS layer separation

Layer	Question	Product	Failure mode
BOIS Core	How should incomplete knowledge become disciplined inquiry and action?	Minimal transition grammar	Core inflation or false universality.
SIMA analysis	What machine appears to be working on this substrate?	Candidate machines, operome, risk map, correspondence map	Diagnosis becomes dogma or self-validation.
BORIS implementation	How is a narrow domain system built and audited?	Roles, registers, protocols, instructions, logs, transition packages	Implementation is mistaken for core.
Experience tests	Does the implementation meet reality?	Observed outcomes, feedback, selection, revised instructions	Documents replace experience.

BORIS as implementation layer and activity physiology

BORIS is the implementation layer of BOIS, not a second core. It turns the grammar into operational organs: value-interface records, unknown registers, protocol cards, instruction cards, evidence levels, experience logs, attention ledgers, role definitions, crisis procedures, transfer protocols, archives, and transition packages. BOIS specifies the transitions; SIMA analyzes the actual and candidate machines; BORIS makes a chosen narrow machine executable and auditable.

The current layer-separation gate is publication-relevant. If BORIS templates, reports, or packages start being treated as BOIS itself, the system loses its main protection: the ability to distinguish grammar, analysis, implementation, and evidence. A scientific article should therefore say that BORIS is an implementation proposal and that any domain-specific BORIS(D,V,S) remains a candidate until tested by experience.

Documents alone are not physiology. An organization may possess many procedures and still be unhealthy if those procedures consume attention, block feedback, or hide out-of-domain cases. In BOIS, physiology is the crystallized capacity of an activity system to act, learn, reproduce useful routines, and release attention for new valuable unknowns. A healthy physiology returns experience into selection and instruction. An unhealthy physiology turns documentation into ritual and automation into the concealment of ignorance.

A human may be modeled functionally as a carrier of attention inside the system, but a human is not reducible to that function. BOIS cannot justify destroying people, rights, safety, or legality in the name of optimization. Attention is a limited operational resource; attention carriers are not disposable objects.

From BOIS to the broader design space

BOIS plus SIMA plus BORIS demonstrates one design, but it does not exhaust the class of philosophical machines. It is transition-oriented: it emphasizes incomplete knowledge, domain boundaries, reversibility, evidence levels, operator authorship, closure quality, preservation of state, and experience-tested physiology. Those virtues have costs. They can slow action, privilege auditability over spontaneity, and fit poorly where the primary task is poetic invention, mass political mobilization, or existential interpretation.

Other machines would have different virtues and failure modes. A skeptical machine maximizes resistance to unsupported claims but may paralyze action. A pragmatic machine privileges reversible experiments and consequences but may mistake usefulness for truth. A scientific-falsification machine demands possible refutation but handles value and meaning poorly when used alone. An ethical machine foregrounds affected parties, consent, dignity, and responsibility, but may slow urgent decisions. A hermeneutic machine preserves context and plurality of meaning but may lose decisiveness. An apophatic machine protects against false naming of what is not yet knowable but can become a refined form of evasion.

The design problem is therefore not to find the most philosophical style. It is to match machine, domain, substrate, risk, and purpose while keeping that match contestable. Composability is possible, but order matters. If ethics is added only after a recommendation is generated, it becomes a filter. If ethics defines the value interface before inquiry begins, it becomes part of the machine. The same modules, arranged differently, create different authority relations.

Substrate: M@S and the physiology of implementation

The broader theory requires a substrate correction: philosophical machines are not disembodied rule sets. A machine must be run by something. The substrate can be a body, committee, software workflow, sensor network, animal collective, institution, robotic swarm, or hybrid human-AI ecology. Once executed, the machine acquires timing, channels, bottlenecks, memory stores, energy costs, failure modes, repair mechanisms, and ethical exposures. The abstract rule may travel; the living implementation does not.

A philosophical machine should therefore be written as M@S: machine M executed on substrate S. Let P(M,S) be the physiology of implementation: the actual mechanism through which M appears in time. The substrate-physiology rule is conditional: if S1 and S2 differ in properties relevant to M, then P(M,S1) and P(M,S2) will differ, even if a selected high-level function is preserved. The rule is too strong if it says that every microscopic material difference changes the relevant function. It is too weak if it says only that hardware differs. Its useful claim is substrate accountability: a design is incomplete until it declares what runs it and what distortions the substrate introduces.

This rule extends the distinction between computational, algorithmic, and implementation levels. It also resonates with work on extended cognition, cognitive artifacts, and epistemic action: thinking often depends on environmental supports. The substrate extension makes this dependence operational for philosophical machines. When the support changes, the machine's practical body changes. The implementation-level language follows Marr's levels, Shannon's concern with channels, and extended-cognition work on external supports (10, 11, 14-16).

Replicators, carriers, bees, and wolves

Richard Dawkins's *The Selfish Gene* is important here not because a gene-centered theory should be imported directly into AI or social philosophy, but because it warns against confusing a propagated pattern with its carrier or survival machine. A rule, norm, meme, protocol, or organizational routine may travel. Each local execution passes through bodies, infrastructures, and selection pressures that change its practical life. The substrate rule applies the same caution to philosophical machines: do not confuse the portable grammar with the vehicle that realizes it. The caution is deliberately drawn from Dawkins's distinction between propagated patterns and carriers (13).

Consider a simple deliberative machine D. It must search under incomplete knowledge, compare options, amplify better evidence, converge without destroying group coherence, and initiate action. On a honeybee swarm, D becomes a physiology of scouts, dances, quorums, thermoregulation, colony timing, and flight. Evidence is not a paragraph or a ballot. It is a dance, a duration, a return, an excitation, and eventually a launch. On a wolf pack, the same abstract D cannot become bee-style quorum dance. It becomes distributed attention, movement initiation, kinship-mediated coordination, vocal and olfactory signaling, bodily risk, hunger, fatigue, injury, and spatial pursuit. Evidence is orientation, scent, movement, social relation, and the changing cost of staying together. The biological examples rely on swarm-intelligence and animal-movement literature, including honeybee nest-site choice and wolf-pack structure (17-22).

The comparison is valid only at a carefully chosen level. Bees and wolves can both search, signal, converge, and move, but the physiology of convergence is different. A swarm is not a committee with wings. A wolf pack is not a parliament with teeth. A language-model workflow is not a biological collective that happens to use tokens. These cases can implement family resemblances of inquiry, choice, or closure, but each physiology must be described separately.

Table 4. One abstract deliberative machine, different substrates

Substrate	Carrier units	Dominant channels	Implementation physiology	Typical failures
Honeybee swarm	Many insects	Dance, pheromone, temperature, local contact	Scout search, recruitment, quorum, thermoregulation, launch	False quorum, environmental stress, colony disruption
Wolf pack	Few mammalian bodies, often kin	Vision, scent, vocalization, movement, social bonds	Attention, pursuit, division of labor, care, conflict management	Injury, hunger, separation, dominance mythology
Human institution	Roles and persons	Speech, documents, law, reputation, procedure	Agenda, evidence, debate, vote, appeal	Capture, bureaucracy, hidden authority, fatigue
AI workflow	Models, prompts, files, interfaces	Tokens, retrieval, logs, tool calls, versioning	Prompting, checking, closure, rollback, transition	Hallucination, context loss, automation bias, weak audit

AI, institutions, and substrate declarations

The same argument applies to artificial and institutional substrates. A committee running D produces agendas, minutes, votes, delays, coalitions, fatigue, reputational costs, and appeals. A software workflow running D produces prompts, logs, tokens, retrieval, versioning, access control, hallucination risk, audit trails, and rollback. A robotic swarm running D produces radio interference, battery limits, collision avoidance, sensor noise, and repair logistics. These are not decorative details. They determine how the machine remembers, forgets, closes, resumes, and fails.

For AI governance, transparency should include more than model provenance and limitations. It should include the declared philosophical machine, the substrate declaration, and the layer map: what runs it, what channels it uses, where memory is stored, what counts as energy or cost, what can fail, who can stop it, what is harmed by wrong optimization, what audit and repair mechanisms exist, which layer is analytical, and which layer is implementation. Human oversight is meaningful only when the human can see what kind of loop and layer they are in.

This reframes alignment. A system aligned to the wrong philosophical machine can be obedient and harmful. It can follow instructions while treating unverified assumptions as facts, outsourcing value judgments, or closing inquiry before affected parties are visible. Alignment must therefore include the choice of the reasoning architecture and the substrate on which that architecture is executed. Current AI-safety and governance debates make this architectural level practically relevant (24-26, 30).

Tests and publication-level discipline

A philosophical machine should be tested at several levels. The internal-coherence test asks whether categories remain distinct under pressure: facts versus hypotheses, protocols versus instructions, verified results versus eternal truth, closure versus forgetting, core versus analysis, and analysis versus implementation. The adversarial-transfer test asks whether the machine fails safely outside its domain. A design built for organized activity should not silently become a theory of consciousness, a legal authority, or a therapist. Transfer requires a domain, value interface, substrate declaration, evidence translation, risk analysis, and stop conditions.

The operational-consequence test asks whether the machine does what it promises. A machine that claims to release attention should reduce rework, clarify responsibility, improve evidence quality, preserve unresolved uncertainty, and return experience into improved routines. If full reporting consumes attention, the report must justify that cost by preserving transition state, lowering future rework, or revealing a valuable unknown. This is the attention-liberation gate introduced by the current core.

Experience tests are not private impressions. They are logged sequences of action, observation, feedback, selection, and update. A domain-specific BORIS is not accepted merely because it is well documented. It must meet reality through tests by experience, while SIMA audits whether the resulting physiology actually matches the intended machine. Slight residual risk may be allowed when it exposes a valuable unknown, but only with control of physiological consequences, stop signals, and rollback or fallback paths.

The closure-quality test asks what kind of ending is produced: final answer, preserved state, list of open questions, rollback path, transition package, or explicit refusal. The substrate tests ask whether each component of M has a carrier in S, how S distorts timing and incentives, and what new forms of harm or responsibility appear when M runs on S. The operome test asks whether the observed opers are sufficiently stable and useful to justify using operome language in an extended analysis.

Publication-level discipline also requires limiting claims. This article is conceptual and methodological. It does not report field validation of BOIS in firms, does not prove that any biological collective is a philosopher, and does not claim that AI systems possess consciousness. Its stronger claim is architectural: systems that govern inquiry should declare their philosophical machine, substrate physiology, analytical layer, implementation layer, and tests by experience.

Table 5. Tests for publication-level evaluation

Test	Question
Internal coherence	Do categories and layers remain distinct under pressure?
Adversarial transfer	Does the machine fail safely outside its domain?
Attention liberation	Does the physiology reduce rework and free attention for valuable unknowns?
Layer separation	Are BOIS, SIMA, BORIS, interface rules, and article claims kept distinct?
Operational consequence	Does it produce the promised organizational or epistemic effect?
Experience test	Does action meet observable feedback and selection rather than documentation alone?
Closure quality	Does closure preserve state, unknowns, rollback, and responsibility when needed?
Mapping test	Does each component of M have an explicit carrier in S?
Distortion test	How does S change timing, incentives, visibility, and error profile?
Ethical test	What new harms or responsibilities appear when M runs on S?

Operome test	Are opers stable and useful enough to justify an operome inventory?
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Research program

A research program can proceed in five steps. First, build a controlled library of philosophical-machine templates: BOIS-like transition machines, skeptical machines, pragmatic machines, scientific-falsification machines, ethical machines, dialectical machines, hermeneutic machines, apophatic machines, and pluralist machines. Second, define stress tests for each template. Third, run them on shared cases: organizational failures, scientific uncertainty, wicked policy dilemmas, situated human-machine action, education, clinical triage, public communication, and software workflow design. Fourth, measure not only answer quality but category discipline, value disclosure, uncertainty preservation, reversibility, substrate distortion, operator agency, and attention liberation. Fifth, use SIMA to reconstruct what machine was actually operating rather than relying only on declarations.

For BOIS itself, the next work should not inflate the core. It should build testable layers: value-interface protocols, evidence levels, attention ledgers, manipulation resistance, crisis mode, transfer protocol, regression tests, SIMA analyses, and field pilots. Only after such pilots can empirical effectiveness be claimed. Until then, BOIS should be treated as a reconstructable philosophical-engineering machine that is internally protected against several common errors and ready for external testing.

For substrate pluralism, the next work is a comparative mapping program. Researchers can ask which components of M survive when moved across substrates, which acquire new organs, which distortions are harmless, and which make the transfer unsafe. Operature correspondence can formalize those comparisons without claiming identity. This turns analogy into design discipline: here is the abstract machine, here is the substrate, here is the physiology, here is the operome, here is the preserved function, here is the distortion, and here is the ethical boundary.

Conclusion

Philosophical machines are not free-floating philosophies. They are architectures of inquiry running through people, documents, institutions, software, bodies, and environments. In the current BOIS/BORIS frame, BOIS is the transition grammar for organized consequential activity under incomplete knowledge, SIMA is the analytical layer that reconstructs candidate machines from substrate physiology, and BORIS is the implementation layer that builds narrow domain systems to be tested by experience.

The update does not make BOIS universal. It makes the conditions of transfer stricter. Business means organized activity in this project only when D, V, S, substrate declaration, SIMA analysis, narrow BORIS implementation, risk controls, and experience tests are specified. Otherwise the term collapses into rhetoric.

The substrate extension strengthens the theory. A philosophical machine should not be specified as M alone, but as M@S. The same abstract grammar can acquire different implementation physiologies on a firm, a committee, a software workflow, a bee swarm, a wolf pack, a robot swarm, or an AI assistant. That difference does not make analogy impossible; it makes declaration, mapping, distortion analysis, ethical testing, and operature correspondence mandatory.

To build accountable AI and accountable institutions, we must declare not only outputs, models, datasets, and values, but also the philosophical machine and the body in which it runs. Hidden philosophy is not neutral; it is merely unaudited. The task is to make these machines inspectable, contestable, limited, transferable only by protocol, and replaceable when their substrate physiology creates unacceptable distortion or harm.

Acknowledgments and disclosures

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